Quantum-inspired Evolutionary Algorithm for Feature Selection in Motor Imagery EEG Classification

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***Abstract*— In Brain-Computer Interfaces, one of the most relevant tasks is the selection of a subset of features that efficiently describes the EEG signal, excluding redundant and irrelevant features. This procedure reduces the dimensionality of the dataset (avoiding the dimensionality curse) and improves the classification accuracy of the system. One of the most successful models applied for this task is the use of an Evolutionary Algorithm in a wrapper approach. These models produce excellent results but present the drawback of a considerable high processing time, a critical limitation for its application on real Brain-Computer Interfaces (BCI) systems. Quantum-inspired Evolutionary Algorithms can be an alternative wrapper approach for the feature selection task, given that they outperform classical Evolutionary Algorithms in the exploration and exploitation of the search space, obtaining the global solution much faster. These algorithm employs concepts and principles from the Quantum Mechanics to probabilistically describe a set of different states between the classical logic states 0 and 1. In this paper, a Quantum-inspired Evolutionary Algorithm is developed and tested over three different subjects from publicly available datasets. In the proposed model, Wavelet Packet Decomposition is employed to analyze the time-frequency characteristics of the signals, and a Multilayer Perceptron Neural Network is employed as a classifier.**

Keywords*— *Quantum-inspired Evolutionary Algorithms; motor imagery; EEG classification; feature selection; Genetic Algorithm

I. Introduction

Quantum Mechanics represents a field of Physics that studies the behavior and properties of matter and energy in an atomic and subatomic scale. Some of its main principles and concepts have served as inspiration for several Computational Intelligence algorithms since the first approach introduced in 1995 [1]. Its applications range from classification [2] [3] tasks to feature selection [2] [4] or signal processing [5].

Some of the most relevant models among these algorithms are the quantum-inspired evolutionary methods. These types of optimization algorithms employ concepts like Quantum Bits (Qubits), coherence, entanglement, and superposition, combined with operations applied to the Qubits known as Quantum Gates. Examples of these models are: Quantum-inspired Evolutionary Algorithms [6] [7] [8] [9], Quantum-inspired Differential Evolution Algorithms [10], Quantum-inspired Particle Swarm Optimization Algorithms (QiPSO) [4], Quantum-inspired Particle Ant Colony Optimization Algorithms [11], among others. There are two main advantages of these methods over their classical counterparts: higher efficiency [7], and an adequate balance between exploration and exploitation, managing to explore the search space with a lower number of individuals and finding a global solution in less time [12].

One of the application fields that can benefit from the application of these models is Brain-Computer Interfaces (BCI). A BCI is a device that employs the translated brain signals to generate control commands for an external agent. Its application has grown exponentially in the last decade, ranging from robotics and assistive technologies to gaming and art devices.

A frequent brain signal acquisition technique employed in BCI is the electroencephalography (EEG), mainly due to being noninvasive and of relatively easy setting. This method employs electrodes placed on the scalp to measure the brain electrical activity produced by a determined cognitive or imagined physical task. Motor Imagery (MI) is a mental task paradigm that produces variations of the sensorimotor rhythms during the imagination of a determined motor task, which can be detected in the EEG of the subject. Depending on the produced amplitude modulation, these variations can be classified into event-related desynchronization (ERD) and event-related synchronization (ERS).

The main stages of an EEG based BCI system can be seen in Fig. 1. Of these stages, feature selection presents high relevancy. The EEG signals produced by the MI tasks are characterized by its low amplitude, low spatial resolution, susceptibility to noise and artifacts, nonlinearity, and temporal variability. Also, the datasets obtained often present a high dimensionality, with a great number of features and few trials. These factors, combined with the high variability of the signal among trials and subjects, highlight the feature selection as an important stage of the BCI system. The feature selection stage provides a subset of features of lower dimensionality and higher relevancy for the classification stage, eliminating redundant and irrelevant features and reducing the total processing time.

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A great number of algorithms for feature selection has been applied in BCI systems, including filters and wrappers approaches [13] [14] [15] [16]. One of the methods that offered better results for this task is the wrapper approach based on evolutionary algorithms, such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and others. Although these models offer a higher classification accuracy compared to other approaches [13], one acknowledged drawback is the extended processing time [17], which is an extremely important limitation for the training phase of every BCI model. In that concern, Quantum-inspired Evolutionary methods can represent a breakthrough, being able to obtain the relevant features in less time, with similar or higher accuracy for the system.

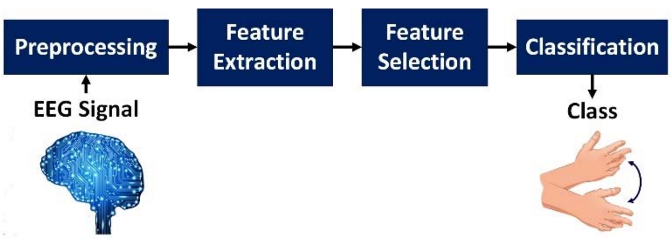

Fig. 1. Stages of a BCI

Despite these advantages, Quantum-inspired algorithms have rarely been applied for feature selection in BCI systems or EEG processing. The only example found in the literature is the QiPSO model [4] [18], which is used in a BCI system with different preprocessing analysis, set of features and classifier than the one proposed in this paper. In this work, the main objective is to develop a Quantum-inspired Evolutionary Algorithm (QiEA) as a variant of a wrapper approach to select the most relevant features for the classification of the EEG data elicited by MI tasks. This EEG data is obtained from channels C3 and C4 of the international standard 10-20 system, with the subject imagining right and left hand movement. The advantages of applying such an approach are evaluated by comparing with a classical GA (best method employed in [13]). The comparison is performed using the classification accuracy obtained from publicly available MI datasets of BCI Competitions and a Multilayer Perceptron Neural Network as the classifier.

The remainder of this paper is organized as follows: next section describes the proposed methods for every functional block of the EEG signal analysis (Preprocessing, Feature Extraction, Feature Selection and Classification), with emphasis on the QiEA theory. Section III presents the MI data used and the results obtained for three subjects with the proposed QiEA and a classical GA. Finally, Section IV provides the conclusions of our work.

II. Materials and Methods

As stated in Fig. 1, a BCI system can be generally represented by four functional blocks. The following sections describe the methods employed for each task, concentrating on the QiEA model.

A. *Preprocessing and Feature Extraction*

The preprocessing stage extracts the essential components from the signal, reducing noise and artifacts, which contributes to an optimal differentiation among the signal classes (cognitive or imagined physical tasks performed by the subject). In this stage, a time-frequency analysis of the signal can be advantageous, coping with the EEG challenging characteristics. By applying this analysis, features can be defined simultaneously in time and frequency domain, dividing the signal into more informative frequency bands without losing temporal resolution. For non-stationary, random, and transient signals, like EEG, the Wavelet Transform is highly recommended for time-frequency analysis, presenting excellent discriminating results [19] [20]. The subsections below give a brief introduction of the techniques employed in the proposed BCI system. More detail is presented in previous works [13].

1) *Wavelet Packet Decomposition*

A particular extension of Wavelet Transform, known as Discrete Wavelet Transform (DWT), is the Wavelet Packet Decomposition (WPD) [21]. This technique allows a highly detailed representation of the high and low frequencies components of the signal, decomposing it into shifted and scaled versions of a mother wavelet through a recursive filtering scheme.

There are two main parameters that determine the successful application of the WPD: the type mother wavelet and the number of levels. The number of levels defines the number of successive filtering steps, establishing the range of each final frequency sub-band. Therefore, the value is selected depending on the characteristics of the signal in the frequency domain (predominant components of the signal in this domain). In the case of the mother wavelet, there exist several models (Daubechies, Symlets, Morlet, among others), and the selection is usually determined by empirical tests.

The mother wavelet chosen was the Daebuchies of order 4 (db4), which is usually employed on EEG signal [22] [20], with four levels, resulting in sixteen sub-bands. From these final sub-bands in the 0 to 64Hz range, with a bandwidth of 4Hz, the ones that contain the main information about frequency components for MI tasks classification are 0-4Hz, 4-8Hz, 8-12Hz, 12-16Hz, 16-20Hz, 20-24Hz, 24-28Hz and 28-32Hz.

The former selection is based on the characteristics of the MI EEG signal. In the BCI community, the major component bands for ERD and ERS are traditionally defined as the mu band (8-13Hz) and beta band (13-30Hz) [23]. As stated in [19], although these bands contain a meaningful part of the signal information, due to the non-stationary characteristic of the EEG, there is a significant variability on the most discriminative frequency bands [24]. This evidence, combined with works reporting an accuracy improvement with a broader spectrum of frequency bands - including delta (from 0.5 to 4Hz), theta (4-7Hz), and gamma (30-40Hz) - [25] [26] [27] [28], led to the selection of a broader frequency scope.

A common procedure after applying the WPD is the direct use of the resultant coefficients as input features. In this work, however, the signals are reconstructed to the time domain for each frequency band. This approach is based on preliminary tests that presented better performance with reconstructed signals than with the wavelet coefficients.

2) Feature Extraction

There are basically four approaches to extract representative features from the WPD [20]: the decomposition coefficients, statistical information of these coefficients, sub-band energies, and transformation modes of coefficients. In the proposed system, in order to generate a diverse set of features, a combination of statistical, power and phase description of the signal is employed. All of them have already been used separately for EEG signals, except the energy ratio of adjacent sub-bands, which was proposed by the authors based on previous works that employed the ratio of other metrics [29] [30]. The parameters of the feature vector are:

- Mean of the absolute values of each sub-band [30].
- Average amplitude change of each sub-band [31].
- Standard deviation of each sub-band [32].
- Energy of each sub-band [33].
- Entropy of coefficients of each sub-band [32].
- Root mean square of each sub-band [31].
- Phase Locking Values of each sub-band [34].
- Energy Ratio of adjacent sub-bands.
- Variance of each sub-band [35].

The calculation of those parameters for each frequency sub-band produces a final feature vector composed of 134 features:

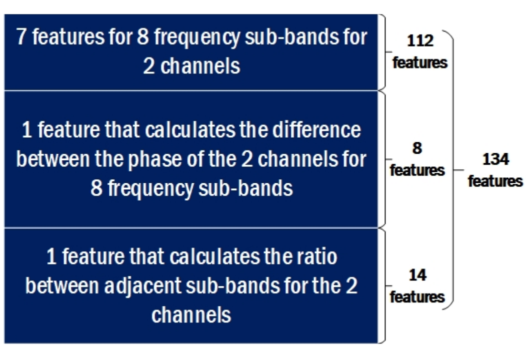

Fig. 2. Feature vector composition

B. Feature Selection

The high dimensionality of the feature vector gives an important role to the feature selection procedure, which helps to obtain an optimal accuracy of the classification model. Feature selection methods are grouped mainly into two categories: filter and wrapper approaches [36]. In the filter approach, the features/subsets of features are evaluated/selected following a certain predefined criterion (independently of the learning model). In the wrapper approach, on the other hand, the criterion for selecting the subset of features is the accuracy of the learning model, making it strictly related to it.

In a previous work [13], which evaluated several feature selection algorithms (including filter and wrapper approaches), the wrapper approach offered a substantially better classification accuracy for the learning model, selecting a subset of features that better describes the dataset for a given classifier (a conclusion also founded in the literature). Despite the high classification accuracy offered by the GA in a wrapper approach employed in [13], this algorithm requires a considerable processing time, which is undesirable for BCI systems due to the need of individual training for each subject. One of the possible approaches that can improve this problem maintaining the classification accuracy is the Quantum-inspired Evolutionary Algorithms, which until our knowledge was never applied for this task on EEG data.

1) Quantum-inspired Evolutionary Algorithms

QiEAs are the result of the integration of Quantum Mechanics concepts and Genetic Algorithms. The efforts done developing these algorithms produced three main categories defined by the type of representation of the chromosome [37]: binary observations QiEAs (best fitted for a wrapper application), real observations QiEAs, and QiEA-like algorithms.

QiEAs with binary representation employ a Qubit as a gene in a chromosome. The Qubits are units that probabilistically represent states $|0\rangle$, $|1\rangle$ or any superposition of both [6]. This representation is defined as a vector of two numbers $[\alpha\ \beta]^T$, complying with the condition that $|\alpha|^2 + |\beta|^2 = 1$. $|\alpha|^2$ and $|\beta|^2$ specify the probability of observing state $|0\rangle$ or $|1\rangle$, respectively. The Qubit state is represented by Equation (1):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

A binary quantum chromosome (q_j^t) is represented as a set of Qubits (consecutive pairs of numbers), as in Equation (2), which are observed in order to obtain classical individuals. This observation procedure is implemented by generating a random number and, if this number is lower than the probability correspondent to the $|0\rangle$ state ($|\alpha|^2$), the classical value is 0, otherwise, the value is 1. The flow chart of the proposed QiEA (based on [8]) is described in Figure 3.

$$q_j^t = \left[\begin{array}{c|c|c|c} \alpha_{j1}^t & \alpha_{j2}^t & \cdots & \alpha_{jm}^t \\ \beta_{j1}^t & \beta_{j2}^t & & \beta_{jm}^t \end{array}\right] \tag{2}$$

Each Qubit of the chromosomes from the quantum population $Q(t) = \{ q_1^t, q_2^t, \dots, q_n^t \}$ is initialized with equal probability amplitude ($1/\sqrt{2}$) for each state $|0\rangle$ and $|1\rangle$. The evaluation stage constitutes the calculation of the fitness function for each classical individual from P(t). The recombination is performed using the uniform crossover [38].

The quantum population is then updated through the application of the quantum rotation gate, defined as:

$$\begin{bmatrix} \alpha_{ji}^{t+1} \\ \beta_{ji}^{t+1} \end{bmatrix} = \begin{bmatrix} \cos\Delta\theta & -\sin\Delta\theta \\ \sin\Delta\theta & \cos\Delta\theta \end{bmatrix} \begin{bmatrix} \alpha_{ji}^{t} \\ \beta_{ji}^{t} \end{bmatrix} \tag{3}$$

where $\Delta\theta$ is the angle that controls the rotation of the values, determining the angular approximation of the Qubit state to $|0\rangle$ or $|1\rangle$. Fig. 4 shows a representation of the rotation gate for a

Qubit. The termination criterion in this case is the maximum number of generations.

To apply this algorithm for feature selection, each Qubit of the chromosome represents one of the extracted features (in our case the chromosome is composed by 134 Qubits). The features with an active value in the chromosome are the ones selected to compose the subset of features. Each subset is evaluated employing the accuracy of a Multilayer Perceptron Neural Network (MLP) as fitness function. The MLP is configured with a fixed topology of 10 neurons in the hidden layer (only one value due to the time constraint imposed by being applied for each observation of the QiEA), and training five models with different random initialization of the weights; the MLP with the best classification accuracy is selected.

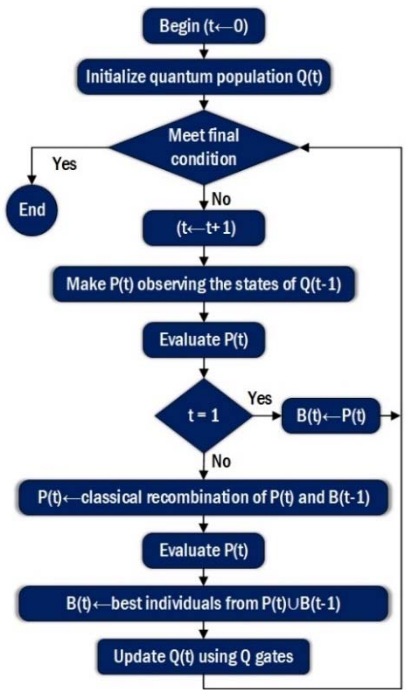

Fig. 3. Flowchart of the QiEA

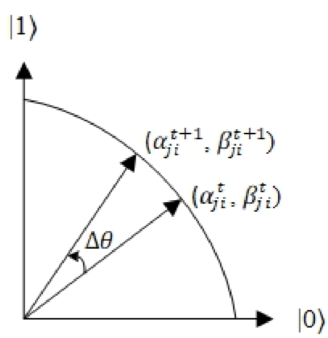

Fig. 4. Polar representation of the rotation gate

The proposed QiEA contains five configuration parameters: size of the quantum population; number of observations to form the classical population; maximum number of generations; recombination rate for the classical individuals; and the update angle.

C. Classification

The model employed for the final classification is an MLP defined with a dynamic topology, varying the number of neurons in the hidden layer between 5 and 20, training 5 networks for each configuration in order to obtain the model with the best accuracy. This classifier has demonstrated to be suitable for BCI applications, presenting high classification accuracy and robustness [39].

III. Results and Discussion

The datasets used in this work were obtained from the BCI Competition II (dataset III-MI) and BCI Competition III (dataset IIIb-MI), offered by the Department of Medical Informatics of the Institute for Biomedical Engineering of Graz University of Technology [40] [41]. The first dataset contains the EEG data obtained from a single subject that performed a two-class MI task (right or left hand movement), and the second dataset contains the measures of three subjects executing the same tasks. Details of the datasets are provided in Table 1. In both datasets, the EEG signals were filtered between 0.5 and 30Hz. From the second dataset, subjects S4 and X11 were selected, since the third one presented fewer trials.

TABLE 1. CHARACTERISTICS OF THE SELECTED DATASETS

Datasets	Trial Time	Electrodes	Sampling Frequency
dataset III-MI	9s	C3, C4	128Hz
dataset IIIb-MI	7s	C3, C4	125Hz

All models were developed using Matlab 2016a software tool. After the preprocessing and feature extraction phases, the dataset consists of 531 trials for S4, 540 for X11 and 280 for A2 (the subject from BCI Competition II database). For the classification process, the dataset was normalized to values between 0 and 1 and divided with 70% trials for training and 30% for test.

The first set of tests was designed with the objective to observe the influence of the configuration parameters on the performance of the QiEA. From preliminary tests, the parameter with the highest impact on the performance outcome was the update angle. To better understand the effects of the update angle, the other parameters were left fixed, with the following values: 40 generations for 5 quantum individuals, 40 observations (8 observations of classical individuals per quantum individual) and 0.8 for the recombination rate. Figure 5 shows the behavior of the algorithm with different values of the update angle for the subject S4. It is relevant to highlight that these results are obtained over a validation set selected from the training set in order to decide the best subset of features.

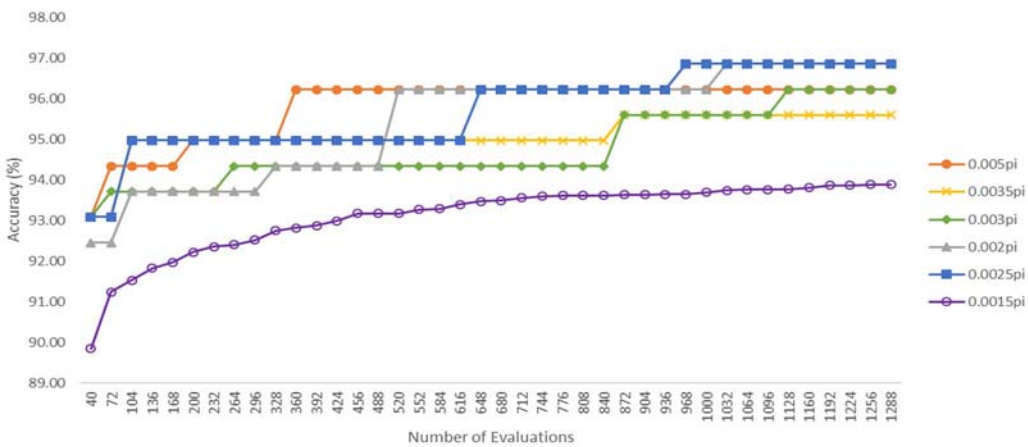

Fig. 5. Graph of the overall accuracy obtained for different update angles

Of all the tested values of update angles, only two reached the global maximum of 96.86% of overall accuracy: 0.002π and 0.0025π. However, 0.0025π managed to reach the value for only 968 evaluations of the fitness function, differently than the 1032 evaluations needed for 0.002π. From the results in Figure 5, it can be verified that a larger value for this parameter (in this case 0.005 π) converges faster to a good result, but not to the global maximum. For smaller values (0.0015π), the results are very poor, not passing the 94% of accuracy.

As mentioned previously, the other parameters are not so critical for obtaining a successful optimization. The number of quantum individuals should be higher or equal to 5, in order to guarantee a good exploration of the search space. In relation to the number of observations, it was noticed that a very high or low number of observations resulted in poorer accuracies, being the best result for 40 observations. The recombination rate provided better accuracy with values over 0.8.

The results obtained with the quantum-inspired evolutionary algorithm were compared to the ones of a standard genetic algorithm employed in a wrapper style, as in [13], with 40 individuals and 40 generations. The accuracy obtained with the standard GA can be seen in Figure 6, with the addition of the best configuration obtained with the QiEA. The graphic reveals the main advantage of the QiEA algorithm, which is the reduction in the number of evaluations needed to reach the maximum accuracy. The GA needed 1440 evaluations to reach 96.86% accuracy, while the QiEA reached that value with only 968 evaluations, which is around 33% less number of necessary evaluations.

The resultant application of the QiEA and classical GA for the other two subjects with the same previous configurations can be seen in Fig. 7 and Fig. 8, respectively. An observation of both graphics confirms the conclusions exposed for the other subject. The QiEA, for both subjects, reached the maximum with an even greater difference of evaluations (632 for X11 and 1008 for A2). This is evidence that the algorithm can effectively obtain the same results as a classical GA, but with less number of evaluations.

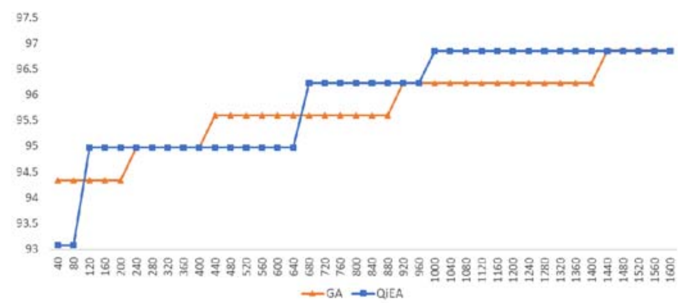

Fig. 6. Overall accuracy obtained by a classical GA

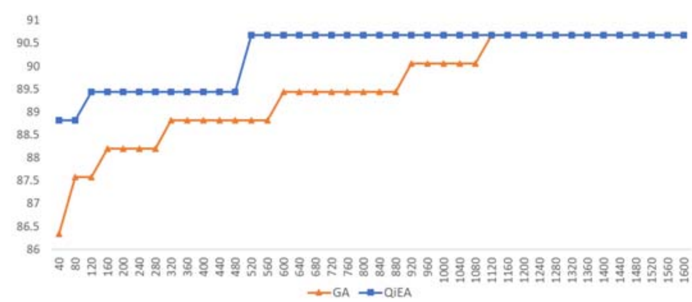

Fig. 7. Accuracies for the QiEA and GA applied to subject X11

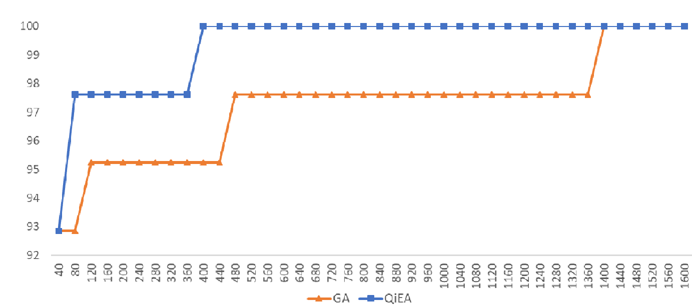

Fig. 8. Accuracies for the QiGA and GA applied to subject A2

Table 2 exposes the overall accuracy obtained from the test set for the subset of features selected for each subject for both QiEA and GA. Also, the processing time of each algorithm and the percentage of processing time that can be saved employing the QiEA are shown, in order to better establish a comparison between them.

In general, the mean accuracy over the test sets maintained similar values, with slightly lower values for the standard deviation for the results of the GA. The interesting result is related to the processing time of the algorithms, with lower values in almost 2000 seconds for the QiEA for subjects S4 and X11, and a more moderated difference (over 1000 seconds) for subject A2. These values represent a reduction of the processing time of 33.84%, 32.58%, and 29.47% for subject S4, X11 and A2, respectively. These processing times represent the processing time for the tested configurations. If the number of evaluations was limited to the values needed to reach the maximum, then the processing time of the QiEA will be very small compared with the one of the GA. These results confirm that the QiEA is faster than the GA and also can offer similar accuracies.

The advantage of the QiEA over the classical GA is the probabilistic representation imbued in the nature of the algorithm. The possibility of representing a linear superposition of states offers a better population diversity than classical GA, which results in the exploration of the search space with a reduced number of individuals. Also, the probabilistically guided evolution allows a better balance between exploration and exploitation in comparison with classical GA.

TABLE 2. RESULTS FOR QIEA AND GA IN THE TEST SETS

	S4		**X11**		**A2**	
	GA	**QiEA**	**GA**	**QiEA**	**GA**	**QiEA**
Mean Accuracy	89.84	89.82	83.74	83.62	95.00	95.29
Standard Deviation	0.98	1.05	0.74	0.89	0.78	0.92
Processing Time (s)	6744	4462	6279	4233	3617	2551
% of time saved by the QiEA	33.84		32.58		29.47	

IV. CONCLUSIONS

This work presented the implementation of a QiEA for feature selection applications on a BCI system based on MI EEG signals. The objective of this model was to improve the processing time of the feature selection stage, maintaining a high performance of the subset of features for each subject. It must be emphasized that the subset of features is subject dependent, demanding a fast training time. The proposed algorithm was applied to three subjects from datasets available at BCI Competition II and BCI Competition III, whose tasks were the imagination of left and right hand movements. The results obtained were compared with the results of a classical GA, both with an MLP neural network as the classifier.

The QiEA greatly outperformed the GA in relation to the convergence time to reach maximum accuracy for each subject. Also, it presented similar accuracy results for the test set, but with much lower processing time. The results confirm that the QiEA is an attractive algorithm for optimal feature selection for BCI applications, offering the advantages of high classification accuracy and faster convergence time when compared to a classical GA.

As future work, the algorithm can be further optimized by implementing new operators as quantum crossover or quantum mutation. Additionally, the whole BCI system can benefit from the use of QiEA with real representation [8], for the evolution of other important parameters, including the ones related to the WPD application or the configuration of the neural network classifier. Finally, the analysis can be extended to other datasets, including our own experimental dataset obtained using the EEG headset Emotiv EPOC+.

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